Decoder-Free Optical Learning for Axial Structure Recovery in Fixed-Focus Fluorescence Microscopy

Supplementary Material

This supplement documents optical training, reconstruction, evaluation protocols and the additional controls supporting the main paper. Axial W1 denotes profile error; earlier control tables labeled depth error use the axial-centroid metric.

S1. Optical implementation and training

Sampling and scan. The optical field uses $0.08\,\mu\mathrm{m}$ lateral and $0.32\,\mu\mathrm{m}$ axial sampling. Analysis sampling is $0.32\,\mu\mathrm{m}$ and reconstruction sampling $0.64\,\mu\mathrm{m}$. The scan uses $\alpha_m = m\pi/16$, $m = 0, \ldots, 15$, and eight uniformly spaced θ values in each of $[\pi/6, \pi/3]$ and $[2\pi/3, 5\pi/6]$. With $\mathbf{s} = \cos\theta(\cos\alpha, \sin\alpha)$, the two intervals supply opposite transverse signs and $\sin\theta > 0$ gives forward propagation. The 256 incident directions are distinct. Phase features have pitch $0.32\,\mu\mathrm{m}$ and the plate distance is $D = 6.6\,\mu\mathrm{m}$ in specimen-equivalent coordinates.

Illuminated field. The simulation extends the plate and illumination aperture to twice the specimen width and crops the resulting excitation to the specimen. The central phase realization is learned while its independently drawn surround remains frozen. This keeps the beam boundary outside the reconstructed field. The plate-free control uses a four-times aperture to reduce boundary-induced modulation. All encoders share the vectorial relay, objective pupil, emission model, pixel averaging and operator assembly.

Dose and counts. Excitation is normalized to a common mean over the full scan and specimen volume. For k directions, the exposure scale is $\kappa_{\mathrm{frame}} = \kappa_0\, 256/k$. The same specimen-specific κ_0 is used across encoders, chosen so the random plate gives 1000 expected photons at the mean pixel signal for $k = 256$. The 256-direction scan has exactly matched deposited excitation; random subsets match in expectation, without subset-specific renormalization. Incident power and detected counts are not assumed identical. Fewer acquired images receive longer exposure. The full-scan mean detected signal differs by less than 4 % across the original untrained overfilled-aperture controls on each test crop. For the learned plates on the 12 prespecified crops, full-scan mean detected signal is about 1–2 % lower than random on average, with per-crop changes from -12 to $+9$ % across the two optimization seeds.

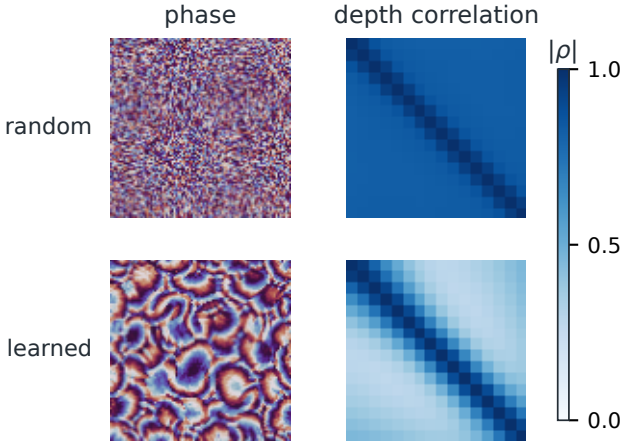


Figure S1. **Learning changes the plate and its depth responses.** Wrapped phase maps and mean absolute depth-signature correlations at 32 directions, before and after training. Correlations average interior lateral positions. The independent Poisson comparison is in Fig. S7.

Training. Each Adam step (learning rate 0.05, 1000 steps) samples 24 lateral positions and four of the 256 directions. The two optimization seeds change this sampling and share the same initial phase. Centered voxel responses are normalized across the sampled angles and detector pixels. The loss gives weight two to valid depth-index pairs (i, j) with $i + j = 14$ on the 16-plane grid, and weight one to other distinct pairs. These are the selected symmetric pairs used by the implementation. Training uses no specimen or decoder.

Signature visualization. The displayed phase is wrapped to $[0, 2\pi)$. The depth-correlation matrices use the 32 directions of measurement draw 0 and average the absolute Gram entries over all interior lateral positions (indices 2–45). The off-diagonal means are 0.845 for the random plate and 0.480 for the learned plate. This is a complete-measurement signature comparison, not a comparison between differently normalized illumination-only and camera-space vectors.

S2. Interpreting the recorded depth signatures

The training statistic centers each voxel response over both angles and camera pixels. This differs from centering an excitation sequence over angles alone. The distinction matters because fluorescence responses have a positive mean.

Let $e_z \in \mathbb{R}_+^T$ be an angular response sequence, with mean

$\mu_z > 0$, population standard deviation σ_z and contrast $\gamma_z = \sigma_z/\mu_z$. If $r_{zz'}$ is the Pearson correlation of two sequences, their mean-kept cosine is exactly

$$c_{zz'} = \frac{1 + r_{zz'}\gamma_z\gamma_{z'}}{\sqrt{(1+\gamma_z^2)(1+\gamma_{z'}^2)}}. \tag{S1}$$

Thus, weak correlation between fluctuations can coexist with high similarity of the non-negative responses when contrast is low. The identity uses each sequence's own contrast; substituting a median contrast does not reproduce the mean pair similarity.

For intuition about the camera-space statistic, approximate co-lateral voxel responses by $q_z = e_z \otimes g$, with a common detection footprint $g \in \mathbb{R}_+^P$. This approximation is appropriate when detection varies little across the axial range. Centering q_z over its TP entries gives

$$\widetilde{c}_{zz'} = \frac{1 - \eta + r_{zz'}\gamma_z\gamma_{z'}}{\sqrt{(1-\eta+\gamma_z^2)(1-\eta+\gamma_{z'}^2)}}, \tag{S2}$$

where $\eta = (\sum_p g_p)^2/(P\|g\|_2^2)$. A localized footprint has small η, so this statistic can be close to the mean-kept cosine. The actual forward model retains depth-dependent footprints and within-voxel variation; it does not use this separable approximation for training or reconstruction.

Preset example. The angular-response panel in the main paper uses the field-center voxel location $(24, 24)$, fixed in the original dump script before visualization, and the 32 directions of draw 0. Each entry is the noiseless response on that voxel's own camera pixel, summed over its fine-grid cells with their emission weights. Rows are scaled by their own maximum only for display. At this location, median angular contrast over depths increases from 0.58 to 1.53, while mean pairwise cosine decreases from 0.83 to 0.41. The mean fraction of directions exceeding 10 % of a depth's maximum decreases from 98 % to 33 %. These describe one preset position, rather than a field-wide mechanism test or an ablation of contrast. The whole-field Gram matrices are computed separately as described in Sec. S1.

S3. Voxel integration and operator assembly

Voxel-integrated operator. To capture excitation variations within reconstruction voxels, we evaluate the optics on a finer grid and integrate the measurement responses over each voxel. Let B be the fine-grid measurement operator, including emission propagation and pixel averaging, and let U replicate each reconstruction coefficient over its fine-grid cells. Then $\mathsf{A} = \mathsf{BU}$ preserves the fine-grid response for every specimen represented in this piecewise-constant basis.

Sub-voxel specimen structure remains unrepresented. For a fine-grid specimen $\mathbf{x}_f$, its coarse representation $\mathbf{x}_{\mathrm{gt}} = \mathsf{R}\mathbf{x}_f$, obtained by voxel averaging, satisfies

$$\mathbf{y} = \mathsf{A}\mathbf{x}_{\mathrm{gt}} + \underbrace{\mathsf{B}(\mathbf{x}_f - \mathsf{URx}_f)}_{\boldsymbol{\delta}_{\mathrm{disc}}} + \boldsymbol{\varepsilon}. \tag{S3}$$

Thus, integration retains within-voxel optical variation, while $\boldsymbol{\delta}_{\mathrm{disc}}$ captures specimen approximation error.

Linear, incoherent fluorescence enables efficient assembly:

$$\mathsf{A}_{(t,p),j} = \textstyle\sum_{i\in\mathcal{P}_j} I_t(\mathbf{r}_i, z_i)\, g_{p,i}, \tag{S4}$$

where $\mathcal{P}_j$ contains voxel j's fine-grid cells, $(\mathbf{r}_i, z_i)$ is the location of cell i, and $g_{p,i}$ is its emission footprint at pixel p, including its integration weight. These footprints are computed once and reused across angles. After precomputation, direct assembly costs $O(TPF)$, compared with $O(TNF\log F)$ for constructing columns through separate FFT-based forward simulations, with T patterns, P detector samples, F fine-grid cells and N reconstruction voxels.

S4. Reconstruction and validation

We evaluate the learned optical encoder using three reconstruction methods: total variation (TV), plug-and-play (PnP) with a pretrained denoiser, and PnP with neuron-trained denoisers (neuron-trained PnP). None is trained jointly with the optical encoder. Each method uses the corresponding forward operator, while its prior remains fixed across encoders. Scalar reconstruction parameters are selected separately for each encoder and compression setting using held-out validation volumes.

Total variation. As a non-learned baseline, we reconstruct the volume using box-constrained least squares with isotropic three-dimensional total variation regularization [6]. The resulting optimization problem is solved using the alternating direction method of multipliers (ADMM) [2].

Multi-slice plug-and-play. PnP reconstruction combines the physical measurement model with denoisers acting as implicit priors [3, 8]. Following multi-slice fusion [4], we denoise a shared three-dimensional volume along complementary orientations. For $o \in \{xy, yz, zx\}$, P_o rearranges the volume into two-dimensional slices, which $\mathcal{D}$ processes independently, giving $\mathcal{D}_o = \mathsf{P}_o^{-1}\mathcal{D}\mathsf{P}_o$.

We combine measurement consistency, denoising and sparsity through PnP-ADMM. The branch set is $\mathcal{J} = \{xy, yz, zx, \mathrm{sp}\}$ for pretrained PnP and includes an additional 3D branch for neuron-trained PnP. With penalties ρ_j, auxiliary volumes $\mathbf{v}_j$ and scaled dual variables $\mathbf{d}_j$, each iteration solves

$$\mathbf{x}^{k+1} = \arg\min_{\mathbf{x}} \tfrac{1}{2}\|\mathsf{A}\mathbf{x} - \mathbf{y}\|_2^2 + \textstyle\sum_{j\in\mathcal{J}} \frac{\rho_j}{2}\|\mathbf{x} - \mathbf{v}_j^k + \mathbf{d}_j^k\|_2^2 \tag{S5}$$

using conjugate gradients. The branches then apply denoising or soft thresholding, followed by

$$\mathbf{d}_j^{k+1} = \mathbf{d}_j^k + \mathbf{x}^{k+1} - \mathbf{v}_j^{k+1}.$$

Pretrained PnP uses a natural-image DnCNN [10] across the three slice orientations and a sparsity branch. *Neuron-trained PnP* replaces this denoiser with a neuron-trained 2D DnCNN shared across orientations and adds a volumetric branch with a neuron-trained 3D DnCNN. Both variants retain the same measurement-consistency formulation. We initialize from normalized back-projection and clip the final estimate to $[0, 1]$. Iteration counts, penalties and parameter-selection procedures are given below.

PnP settings. Both PnP decoders run a fixed budget of 20 ADMM iterations, each with at most 20 conjugate-gradient iterations (relative tolerance 10^{-5}) for the main paper's volume update, start from the normalized back-projection and clip the result to $[0, 1]$. The base penalties are $\rho = 5$ for the denoising branches and 10 for the sparsity branch, both multiplied by one selected scale s; the three slice branches have weight 0.2, the additional 3D branch weight 1, and the sparsity branch $0.2\,l$. Zero-weight branches are omitted. The scales are selected per encoder and compression by mean validation PSNR on $s \in \{\frac{1}{16}, \frac{1}{8}, \ldots, 4\}$ and $l \in \{0, \frac{1}{16}\}$; a selection on an edge extends the grid (s halved or doubled, l multiplied by four), at most twice, and no reported selection remains on an edge. The pretrained denoiser is SCICO's 17-layer DnCNN [1] with frozen parameters; the neuron-trained denoisers are blind, trained over $\sigma \in [0, 0.25]$, and receive no noise level. The branch penalty therefore sets the effective denoising strength.

Data and training. Training volumes come from SOM009 (cell 1, stacks 8 and 9), an animal that contributes no test cell; stack 10 is held out to monitor denoising. Planes whose mean intensity drops below half the median of their neighbors, or repeat the previous plane, are excluded. 459 training crops are cut on the $16 \times 48 \times 48$ reconstruction grid with the preprocessing of the test volumes. The 2D DnCNN (17 layers, 64 channels) is trained on random xy, xz and yz slices for 30k steps (batch 64), the 3D DnCNN (12 layers, 48 channels) on whole volumes for 15k steps (batch 8), both with Adam (10^{-3}), $\sigma \sim \mathcal{U}(0, 0.25)$, flips, lateral transposes and intensity scaling in $[0.3, 1]$; on held-out crops they gain 16.6 and 16.8 dB at $\sigma = 0.1$.

TV and parameter selection. TV uses a box constraint throughout optimization, whereas PnP clips only the final estimate. Scalar parameters are selected separately for encoder and measurement condition on the two vascular validation crops. TV uses 300 ADMM iterations with unit penalties for the TV and box branches, and up to 20 CG steps per update at relative tolerance 10^{-5}. Its initial weight grid is $\{0.01, 0.03, 0.1, 0.3, 1, 3\}$; edge optima extend it by factors of three, at most twice. PnP selects a penalty scale and sparsity scale. Edge selections extend the grid before test evaluation. Test crops are not used for selection. The neuron-trained configuration includes both the three 2D slice branches and the additional 3D branch; it is not a 3D-only replacement.

Computation. On an RTX 4500 Ada GPU, 1000 optical-training steps take about 20 minutes, operator assembly six minutes, and a $C = 8$ reconstruction 4–6 seconds with PnP or 17 seconds with TV.

S5. Axial-profile evaluation

Metric. Axial Wasserstein-1 compares normalized non-negative column profiles [5]. For column c, write $p_{c,\ell} = x_{c,\ell} / \sum_m x_{c,m}$ and $\hat{p}_{c,\ell} = \hat{x}_{c,\ell} / \sum_m \hat{x}_{c,m}$, with cumulative sums $F_{c,\ell}$ and $\hat{F}_{c,\ell}$. Its discrete value is $\Delta z \sum_\ell |F_{c,\ell} - \hat{F}_{c,\ell}|$. We average using reference column brightness over columns above 10 % of its maximum, with $\Delta z = 0.64\,\mu\mathrm{m}$. Numerical normalization uses a 10^{-12} denominator stabilizer; none of the evaluated reference-selected columns had zero reconstructed intensity. For unit-mass profiles, W1 is at least the absolute centroid difference and is zero only when the profiles coincide. It evaluates depth distributions, not connectivity or total brightness.

Re-evaluation. W1 was added without retraining the plate or denoisers. Re-evaluations retain the recorded angular subsets, noise draws and validation-selected parameters. W1, centroid error and image quality in the new tables are taken from the same reconstructed volumes; earlier control tables retain recorded centroid metrics. These runs are not bit-identical replays. TV met the initial per-row tolerance of 0.05 dB and $0.005\,\mu\mathrm{m}$; the PnP re-evaluations did not. The later sweep checked the first row of each configuration within 0.3 dB and $0.02\,\mu\mathrm{m}$, without gating later rows. Table S1 gives the largest deviations, including subsequent rows. The strict TV layered-column result and the PnP re-evaluations are distinguished below.

Full test set. Table S2 reports the 60-crop comparison. Draws are averaged per crop before averaging crops. Cell-bootstrap samples retain all crops from each selected cell; confidence intervals concern the paired relative change of the means. The second optical seed changes the sampling during plate optimization, with the initial phase fixed. The W1 compression sweep uses the same random initial plate.

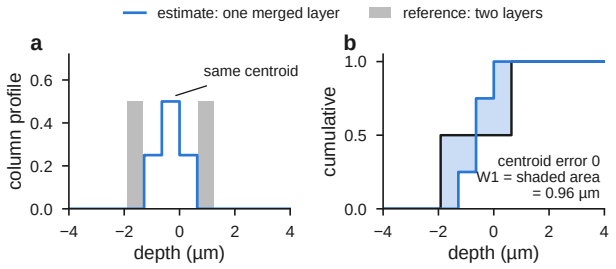


Figure S2. **A centroid can be correct while the axial profile is wrong.** Synthetic illustration: two layers and one merged layer share a centroid. Their W1 is the area between the cumulative profiles. This panel explains the metric; it is not a reconstruction result.

group	rows	max. $\lvert\Delta\text{PSNR}\rvert$	max. $\lvert\Delta\text{centroid}\rvert$
initial profiles	2100	0.198	0.0204
main	1020	0.091	0.0353
objectives	864	0.259	0.0117
dose	1800	0.306	0.0216
compression	552	0.210	0.0061
tolerance	1104	0.220	0.0252
diffusion	240	0.000	0.0000

Table S1. Re-evaluation differences from the earlier recorded reconstructions. PSNR differences are in dB and centroid differences in μm. The initial-profile group includes the strictly checked TV and descriptive PnP evaluations. The reported maxima include rows after the first-row checks.

Neuronal example selection. We rank all 60 crops by the mean fractional W1 change across the three priors, after averaging the five draws, and show ranks 15, 30 and 45. The crop rule was fixed before viewing the images. The selected crops are SOM006_cell1_x1, SOM006_cell3_x5 and SOM005_cell1_x4; every panel uses draw 0. Section and profile locations use reference structure: among interior rows and columns 2–45 with reference peaks away from the first and last planes, the section maximizes the number of qualifying layered columns, with reference energy breaking ties. The profile column has the largest reference signal among the qualifying columns. Boundary exclusions were refined during figure development and applied to all three cases.

Display. Lateral projections use a common linear range from zero to the reference crop's 99.5th percentile. Axial views use fixed 16-voxel lateral windows around the selected profile column, shifted inward at crop boundaries. For these windows, the shared range ends at the reference window's 99.5th percentile. All reconstructions share the reference range for that view; there is no smoothing, sharpening or per-panel adjustment. The window has the physical aspect ratio of its $10.24 \times 10.24\,\mu$m extent. Locators mark its position, and the gold line in each supplementary section marks the profile column. Profile curves are normalized to unit mass and share axes within each crop. Some selected columns worsen after learning even when whole-crop W1 improves; none of these neuronal sections demonstrates separation of stacked neurites.

Layered columns. A signal column is layered when its reference profile has at least two peaks, each with height and prominence at least 20 % of the column maximum. Of the 60 crops, 59 contain such columns. Under the strict TV reproduction check, their W1 falls from 0.467 to $0.309\,\mu$m, a 33.8 % reduction (95 % CI: 31.7–36.3 %). The relative reduction is not larger than the corresponding centroid improvement. Peak recovery remains incomplete: overall, the learned and random plates recover both outer layers in 12 versus 4 % of layered columns with TV, 21 versus 12 % with pretrained PnP, and 20 versus 13 % with neuron-trained PnP. These peak-based summaries are descriptive.

decoder	plate	W1	centroid	PSNR	FG-SSIM
TV	random	0.371	0.259	36.70	0.850
TV	learned s0	0.218	0.150	38.29	0.910
TV	learned s1	0.218	0.149	38.31	0.910
pretrained PnP	random	0.579	0.490	35.72	0.806
pretrained PnP	learned s0	0.290	0.224	37.56	0.900
pretrained PnP	learned s1	0.289	0.222	37.58	0.900
neuron-trained PnP	random	0.271	0.197	38.01	0.902
neuron-trained PnP	learned s0	0.195	0.139	39.60	0.934
neuron-trained PnP	learned s1	0.194	0.136	39.64	0.934

Table S2. Reconstruction metrics from the same re-evaluations ($C = 8$, 60 crops, five draws). W1 and centroid error are in μm; PSNR is in dB. Neuron-trained PnP uses the combined neuron-trained 2D and 3D denoising configuration.

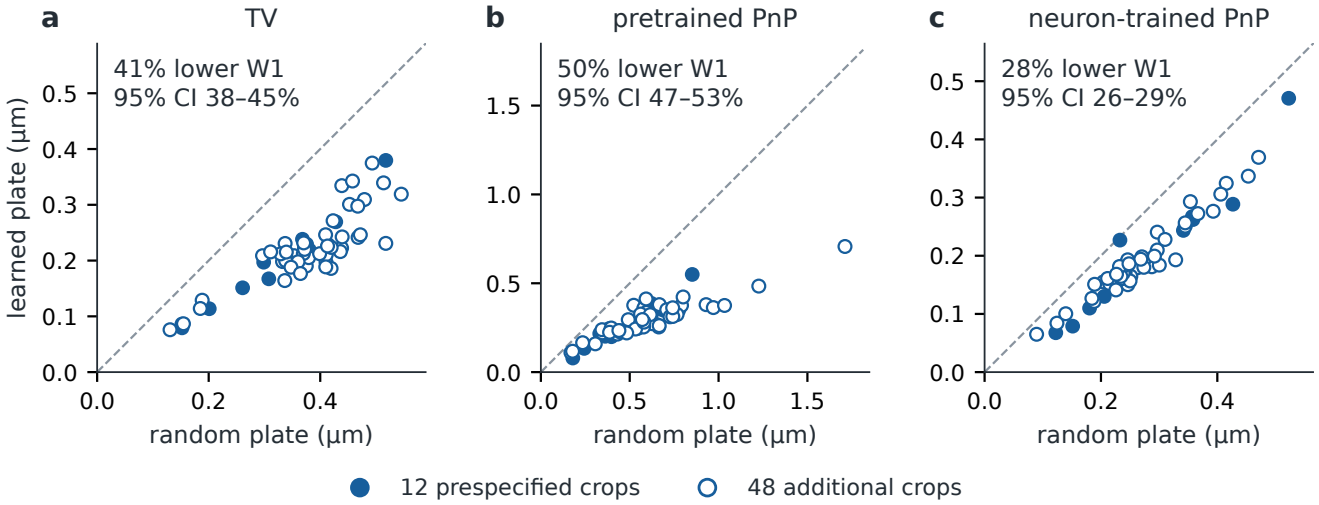


Figure S3. **Crop-level axial-profile recovery.** Each point averages five draws at $C = 8$ and matched nominal dose. The equality line compares learned and random W1; filled and open markers distinguish 12 prespecified and 48 additional crops. All 60 improve under each method. The horizontal and vertical scales agree within each panel.

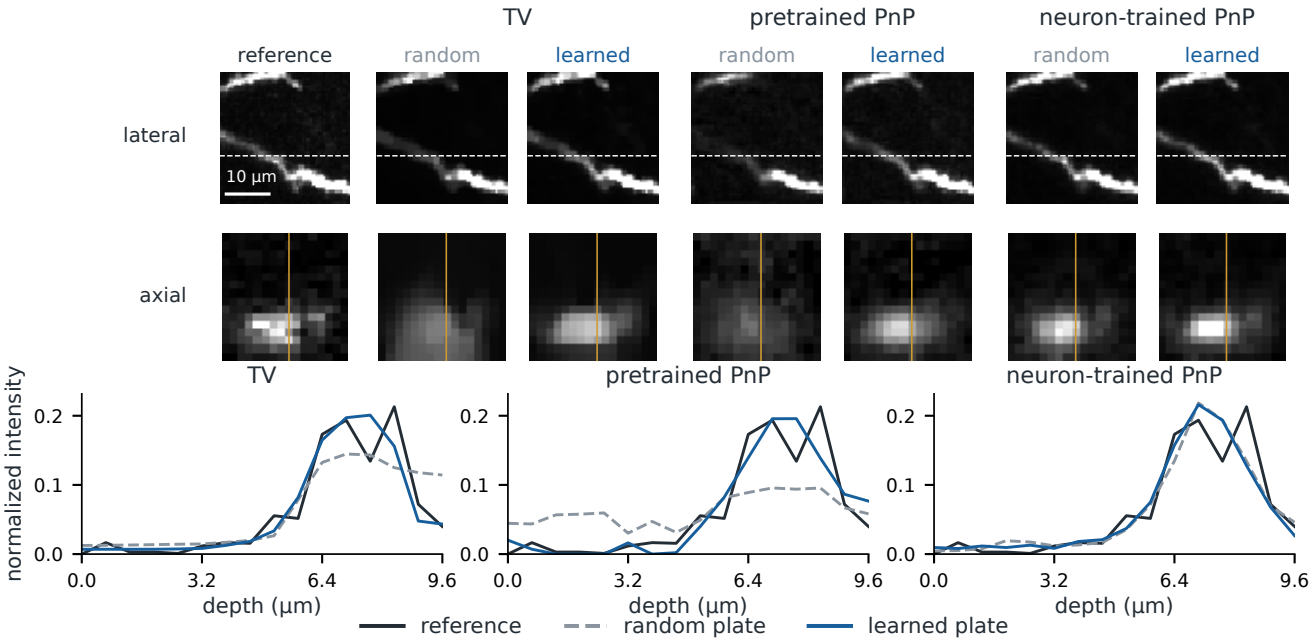


Figure S4. **Rank-15 neuronal example.** Reference and reconstructed lateral projections (top), the marked axial section (middle), and unit-mass profiles at the gold line (bottom). Corresponding views share image scales, and profile axes agree within the crop. Random and learned denote the plates; the priors remain fixed. This case shows reduced smearing, especially under TV and pretrained PnP. At the marked column, neuron-trained PnP has slightly higher profile error after learning despite lower whole-crop W1.

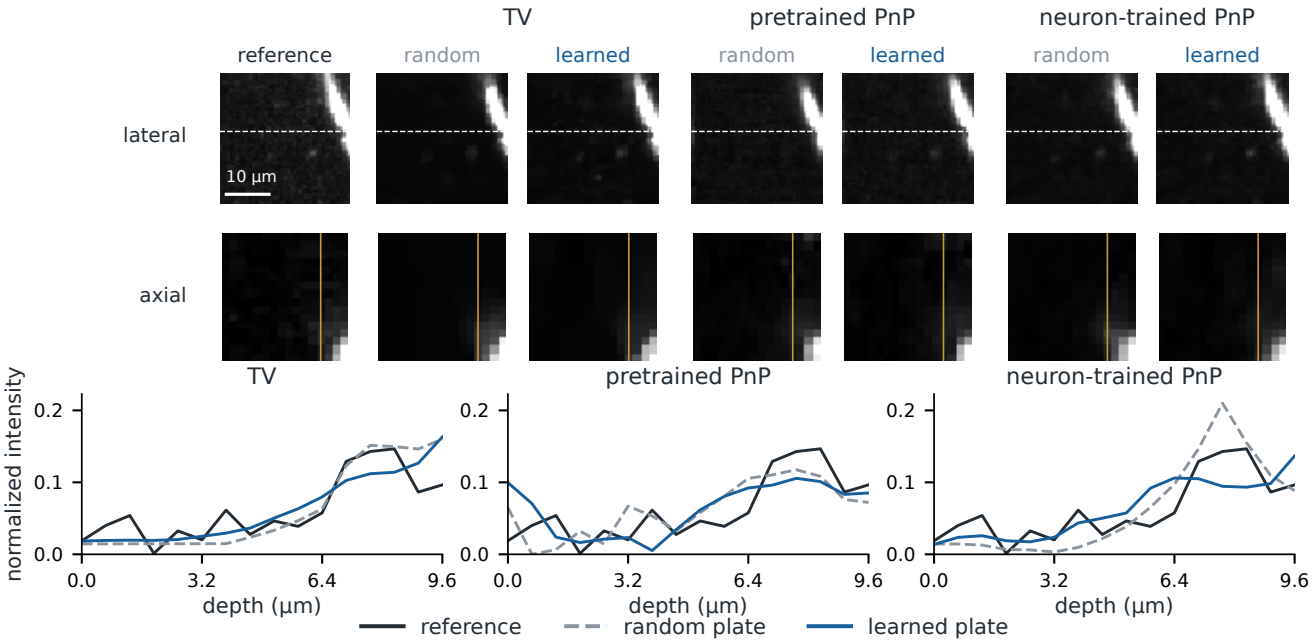


Figure S5. **Rank-30 neuronal example.** Display conventions match Fig. S4. Visible differences are small in the marked section; the pretrained-PnP profile at this column is less accurate with the learned plate despite lower whole-crop W1. The small faint structure remains weak under all reconstructions.

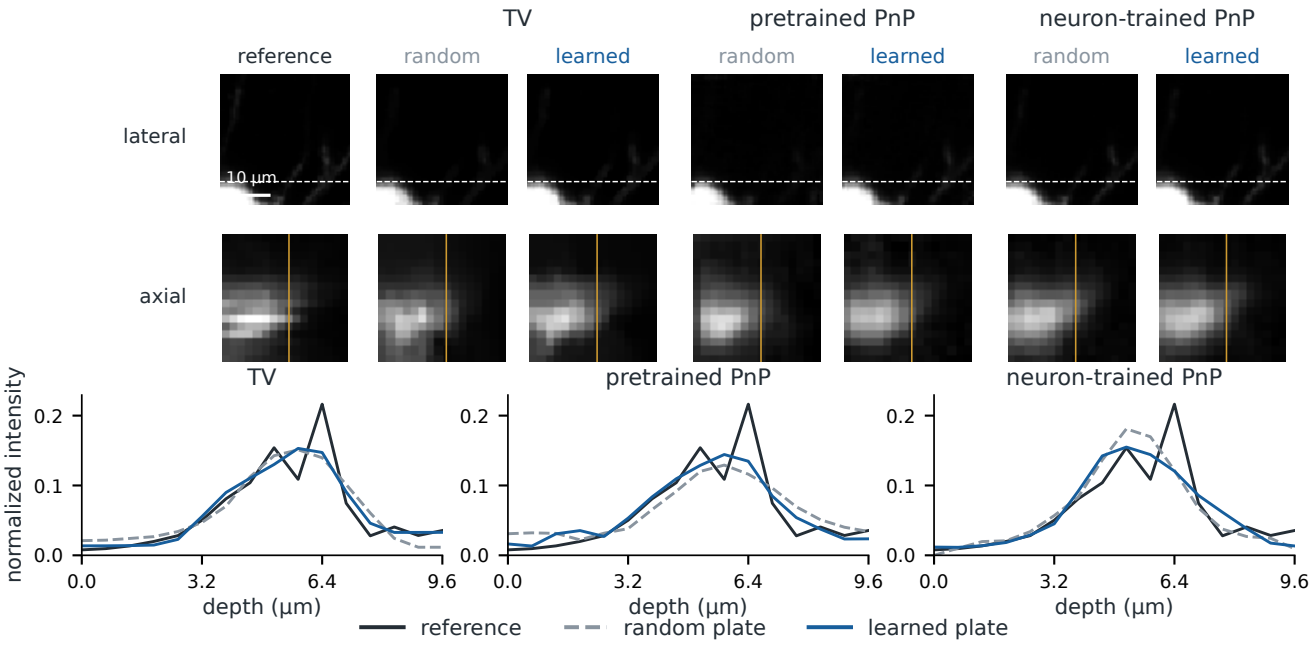


Figure S6. **Rank-45 neuronal example.** Display conventions match Fig. S4. This lower-gain example has modest changes in the section and profiles. The three examples are retained by the ranking rule rather than selected for the strongest visible difference.

S6. Measurement distinguishability and crossing structures

Merged-depth alternatives. In each layered reference column, we move the fine-grid intensity to its axial centroid, distributing mass between the two neighboring planes when needed. This preserves the lateral projection and centroid while changing the depth arrangement. Other columns remain unchanged. The comparison uses the fine-grid forward model, not its piecewise-constant reconstruction approximation. One crop has no layered columns and is excluded, leaving 59.

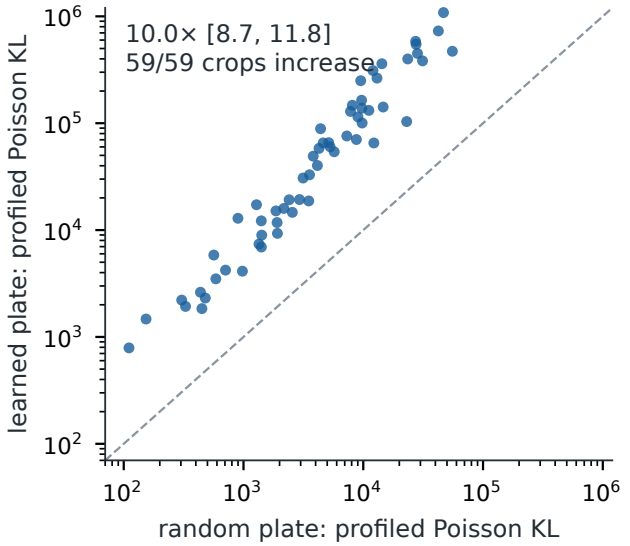


Figure S7. Brightness-profiled Poisson separation between each neuronal reference and its merged-depth alternative. Each point averages five subsets; the factor and interval summarize paired crop ratios. All 59 eligible crops increase.

Poisson comparison. Let s and s' be the expected signal-count vectors of the reference and alternative, and let b be the known background. With $\mu = s + b$ and $\nu(\beta) = \beta s' + b$, we compute

$$\min_{\beta>0} \sum_i \left[\mu_i \log \frac{\mu_i}{\nu_i(\beta)} + \nu_i(\beta) - \mu_i \right]. \tag{S6}$$

At zero background the optimum is $\beta = \sum_i s_i / \sum_i s'_i$. If $p = s/\sum_i s_i$ and $q = s'/\sum_i s'_i$, the profiled divergence equals $(\sum_i s_i) D_{\mathrm{KL}}(p\|q)$. It therefore compares the expected count patterns after removing a global brightness difference. We average over the five angular subsets for each crop and report the geometric mean learned/random ratio with a cell-bootstrap interval. With background, brightness is fitted numerically to the signal while the background stays fixed.

The primary ratio is 10.0 [8.7, 11.8], and every eligible crop improves. Added background gives 10.4 [9.0, 12.4]. Random-plate divergences are already substantial for some alternatives; this result measures increased separation rather than proving that an alternative is measurement-consistent. Swapped-depth, mirrored-depth and equal-W1 rigid-shift alternatives on phantoms provide additional comparisons. The improvement is not uniformly larger for merging than for a rigid shift of matched transport distance, so we make no preferential structural-versus-centroid claim.

Crossing phantoms. Two Gaussian filaments of width $\sigma = 0.32\,\mu\mathrm{m}$ cross at the lateral field center. The three orientation/crossing-angle pairs are $(0°, 90°)$, $(30°, 60°)$ and $(60°, 45°)$. Axial centers are reconstruction-plane indices 7 and 5; only the former coincides with the mirror

pairing used by the plate objective. Gaps are 0, 2, 3, 4, 6 and 8 planes, with $0.64\,\mu\mathrm{m}$ per plane. The zero-gap case is a merged control. We use five measurement draws, the same photon protocol as the neuronal volumes and the existing validation-selected reconstruction parameters.

Success rule. Crossing columns contain both reference filaments and resolve their peaks. Estimated peaks must match the two reference depths within one plane, with height and prominence at least 20 % of the column maximum. At least half of the reference crossing-column brightness must satisfy this condition. Correct ordering additionally requires the estimated intensity centroids along the two separate filament arms to follow the reference depth order, with a margin greater than half a plane. Arm columns contain at least 20 % of their own filament's maximum projected intensity and less than 2 % of the other's. A volume succeeds only when both tests pass.

The reported threshold requires at least 80 % success among the 15 geometry/draw combinations at each axial center, at that gap and every larger tested gap. TV and pretrained PnP improve from 6 to 4 planes; neuron-trained PnP remains at 6. Neither plate produces false splitting in the merged control. The main figure pools the two centers for its curves; the threshold is checked at each center separately. Unequal-brightness filaments use a weaker branch of amplitude 0.3. Closely spaced three-filament crossings remain difficult, and the selected-alternative KL does not by itself guarantee their recovery. Pooled KL/success associations are confounded by gap and phantom type and are not used as a predictive guarantee.

S7. Profile error under acquisition and model changes

Dose and noise. Tables S4 and S5 give W1 over the 12 prespecified crops. Dose-dependent reconstruction parameters are selected on the validation volumes using the same grid-extension rule. At 0.3 times dose the learned plate improves on the random plate at full dose under all three methods. Dose levels 0.1 and 3 provide the remainder of the curve. A plateau of these curves is a property of the tested inference protocols, not proof that additional photons could never help another reconstruction method.

Optical objectives. The W1 comparison across both seeds and all three decoders is in Tab. S6. It uses the same optical initialization and optimization budget. The penalty families are described in Sec. S8.

Alignment. Table S7 reports W1 under alignment errors. Rebuilding the operator for the displaced plate retains the gain. Using the nominal operator under axial or lateral displacement eliminates it; the one-feature lateral shift makes the learned plate worse on W1. Pointing errors retain a smaller advantage. This comparison supports the need for calibration, not unconditional alignment robustness.

decoder	plate	$C = 1$	$C = 4$	$C = 8$
TV	random	0.249	0.298	0.318
TV	learned	0.170	0.208	0.196
pretrained PnP	random	0.413	0.445	0.496
pretrained PnP	learned	0.222	0.254	0.277
neuron-trained PnP	random	0.203	0.234	0.282
neuron-trained PnP	learned	0.142	0.167	0.206

Table S3. Axial W1 (μm) versus compression on the 12 prespecified crops at matched nominal total dose. Fewer directions receive longer exposure per image.

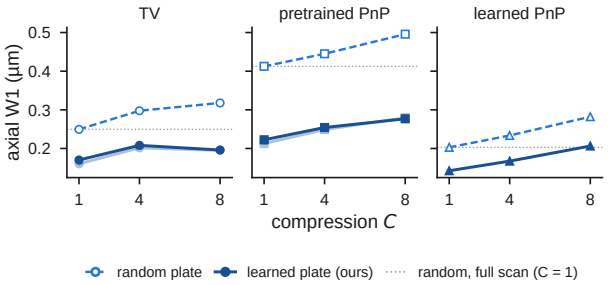


Figure S8. Learned and random plates under fewer illumination directions. W1 is evaluated on the 12 prespecified crops. The horizontal reference is each decoder's random-plate result at $C = 1$.

decoder	plate	$0.1\times$	$0.3\times$	$1\times$	$3\times$
TV	random	0.466	0.442	0.318	0.315
TV	learned	0.292	0.234	0.196	0.192
pretrained PnP	random	0.834	0.631	0.496	0.494
pretrained PnP	learned	0.407	0.302	0.277	0.276
neuron-trained PnP	random	0.345	0.304	0.282	0.275
neuron-trained PnP	learned	0.239	0.220	0.207	0.203

Table S4. Axial W1 (μm) versus deposited excitation dose at $C = 8$. Parameters are reselected on validation data for each condition.

condition	decoder	random	learned	reduction [95 % CI]
background	TV	0.324	0.227	30 % [22, 38]
background	pretrained PnP	0.627	0.297	53 % [48, 57]
background	neuron-trained PnP	0.288	0.210	27 % [19, 35]
0.1× dose	TV	0.466	0.292	37 % [29, 45]
0.1× dose	pretrained PnP	0.834	0.407	51 % [45, 57]
0.1× dose	neuron-trained PnP	0.345	0.239	31 % [24, 37]

Table S5. Axial W1 (μm) with background/read noise or reduced dose. Reductions compare paired means; intervals use cell resampling.

comparison	decoder	W1 reduction [95 % CI]	improved
coherence, s0	TV	17 % [15, 20]	12/12
coherence, s0	pretrained PnP	20 % [16, 25]	12/12
coherence, s0	neuron-trained PnP	13 % [12, 15]	12/12
coherence, s1	TV	16 % [14, 19]	12/12
coherence, s1	pretrained PnP	17 % [13, 23]	11/12
coherence, s1	neuron-trained PnP	13 % [11, 15]	12/12
Fisher, s0	TV	14 % [12, 16]	12/12
Fisher, s0	pretrained PnP	19 % [17, 22]	12/12
Fisher, s0	neuron-trained PnP	14 % [12, 17]	11/12
Fisher, s1	TV	14 % [12, 16]	12/12
Fisher, s1	pretrained PnP	16 % [12, 20]	12/12
Fisher, s1	neuron-trained PnP	16 % [14, 18]	12/12

Table S6. W1 reduction of our plate against each alternative objective ($C = 8$, 12 prespecified crops). Seeds differ in optimization sampling. Counts refer to crops with lower W1.

condition	random	learned	change [95 % CI]
updated model, D=6.1	0.285	0.205	−28 % [−35, −22]
updated model, D=7.1	0.284	0.209	−27 % [−34, −20]
nominal model, tilt 0.005	0.318	0.254	−20 % [−25, −16]
nominal model, tilt 0.01	0.379	0.328	−13 % [−16, −10]
nominal model, D=6.1	0.563	0.586	+4 % [+0, +8]
nominal model, D=7.1	0.447	0.473	+6 % [+5, +7]
nominal model, one-feature shift	0.451	0.500	+11 % [+2, +23]

Table S7. Alignment sensitivity in W1 (μm, neuron-trained PnP, $C = 8$). Changes compare learned against random under the same perturbation. Updated-model rows rebuild the operator and reselect parameters on validation.

S8. Alternative optical objectives

All designs share the phase initialization, sampled positions/directions and 1000-step Adam budget. The coherence alternative uses a smooth maximum of absolute correlations over depth pairs. It belongs to the coherence-design family, but is not a reimplementation of Miniscope3D's shifted PSF cross-correlation and lateral-transfer objectives [9]. The Fisher-inspired alternative uses a discrete, stabilized depth-information surrogate with brightness treated as a nuisance parameter [7]. The uncentered alternative removes column centering from the proposed squared-correlation loss.

	axial W1 (μm) ↓		ours vs this
objective	$C = 1$	$C = 8$	reduction [95 % CI]
random plate	0.203	0.282	27 % [20, 34]
worst-case coherence	0.172	0.238	13 % [12, 15]
depth-Fisher surrogate	0.164	0.240	14 % [12, 17]
decorrelation, uncentered	0.142	0.207	0 % [−1, 1]
depth decorrelation (ours)	0.142	0.206	–

Table S8. Absolute W1 (μm) for the optical-objective comparison under neuron-trained PnP. The final column gives the W1 reduction with our plate against each row at $C = 8$.

Implemented criteria. For normalized depth correlations $\rho_{zz'}$, the coherence loss uses $\frac{1}{50}\log\sum_{z\neq z'}\exp(50|\rho_{zz'}|)$, averaged over sampled positions. The proposed loss divides its weighted sum by $\sum_{z\neq z'} w_{zz'}$ and averages over positions and sampled angle sets.

For the Fisher-inspired criterion, let R be the raw sampled response tensor, $m = \max(\text{mean } R, 10^{-30})$ and $s = R/\operatorname{stopgrad}(m) + 10^{-3}$. The mean is recomputed per update and held constant for differentiation. Finite differences d use unit plane spacing, central differences in the interior and one-sided differences at the boundaries. The criterion uses

$$F_z = \sum_i \frac{d_i^2}{s_i} - \frac{(\sum_i d_i)^2}{\sum_i s_i}, \qquad \mathcal{L}_F = -\mathbb{E}\log\max(F_z, 10^{-12}). \tag{S7}$$

Thus, this is the tested computational surrogate, with its stated scaling, stabilization and gradient convention; it is not a comparison against all Fisher-information optical-design methods.

The two-seed, three-prior comparisons are in the main paper and Tab. S6. Under neuron-trained PnP, the uncentered and centered losses give nearly equal W1 (0.207 and 0.206 μm at $C = 8$). The evidence supports the tested finite-depth criterion, rather than attributing the gain to centering.

S9. Optical controls and axial encoding

The random phase plate, defocused binary amplitude mask, plate-free angular scan and in-focus DMD use the same relay and detection model. The amplitude mask has 50 % open 0.32 μm features at the plate plane. The DMD uses independent binary patterns with 50 % on pixels imaged at the illumination image plane. The phase and mask comparisons include three random realizations; the main learned/random comparison uses realization 0.

The random plate produces approximately linear depth-dependent lateral shifts with a pivot near the plate's conjugate plane. Its excitation sequences decorrelate over roughly 1 μm of axial separation. For point emitters with unknown brightness and 32 illumination directions, maximum-likelihood depth misassignment is 1.5–1.7 % for random

		multi-slice PnP			box-constrained TV			beads (PnP)	
C	encoder	PSNR	FG-SSIM	depth	PSNR	FG-SSIM	depth	depth	ghost
1	phase plate	35.73	0.814	0.357	38.15	0.881	0.187	0.00	0.033
	amp. mask	35.24	0.788	0.347	37.59	0.857	0.191	0.00	0.030
	no plate‡	30.87	0.538	0.753	32.78	0.617	0.477	0.32	0.363
	in-focus DMD	28.96	0.501	2.168	28.48	0.435	2.285	3.84	1.007
4	phase plate†	35.40	0.801	0.389	37.34	0.852	0.223	0.01	0.013
	amp. mask†	34.88	0.778	0.382	36.79	0.823	0.239	0.01	0.017
	no plate‡	30.41	0.513	1.036	32.09	0.580	0.706	0.34	0.452
	in-focus DMD	28.91	0.493	2.168	28.49	0.465	2.326	2.94	1.013
8	phase plate†	34.86	0.784	0.430	37.00	0.847	0.243	0.01	0.039
	amp. mask†	34.14	0.744	0.446	36.74	0.827	0.237	0.08	0.048
	no plate‡	30.18	0.494	1.102	31.81	0.579	0.737	0.49	0.335
	in-focus DMD	28.83	0.482	2.169	28.53	0.473	2.311	2.60	1.013

Table S9. Untrained encoder controls on the 12 prespecified crops, matched nominal dose. Depth error is the recorded axial-centroid metric (μm), not W1. PnP phase/mask entries average three realizations; TV uses realization 0. The plate-free control uses a four-times aperture. The depth-blind centroid references are $1.32\,\mu\mathrm{m}$ (global-profile oracle) and $2.15\,\mu\mathrm{m}$ (uniform depth).

phase plates, 4.4–5.3 % for defocused amplitude masks, 25 % for the tested DMD and 89 % without a plate. The learned plates give rates of 0.38 % and 0.66 % for the two optimization seeds. These diagnostics concern the implemented acquisition, not a universal property of amplitude encoders.

The amplitude mask is competitive in centroid error, while the phase plate improves PSNR by 0.4–1.2 dB at $C = 8$. The DMD stays close to the uniform-depth reference under the tested priors. This is an in-focus control, not a claim about DMD systems generally.

encoder	k	cos(adjacent)	cos(mirror)	axial CRB (µm)	mirror d'	wrong depth (%)	mirror (%)
phase plate	256	0.940	0.823	0.134	10.2	1.0	0.01
	32	0.943	0.828	0.137	10.2	1.6	0.04
no plate	256	1.000	0.999	3.373	0.8	88.2	4.10
	32	1.000	0.999	3.376	0.7	88.7	4.25
in-focus DMD	256	0.967	0.990	0.183	2.4	24.6	13.23
	32	0.970	0.991	0.185	2.4	25.1	13.21
defocused amplitude mask	256	0.966	0.904	0.171	8.0	3.1	0.06
	32	0.969	0.908	0.180	7.7	4.5	0.09

Table S10. Point-emitter diagnostics on the overfilled geometry: detector-column similarities, axial Cramér–Rao bound, mirror separation and wrong-depth rates. Depth precision and emitter misassignment are not optical-resolution measurements. The diagnostic uses the Gaussian approximation to the reference shot-noise exposure and unknown brightness.

S10. Noise and registration protocols

Background and read noise. Background equals the random plate's mean signal per pixel and image, using the same reference for all encoders, with independent Gaussian read noise of two electrons. Its known mean is subtracted before decoding. The reduced-dose condition uses one tenth of the reference exposure and shot noise only. Reconstruction parameters are reselected for each condition on validation data with the same grid-extension rule; no test-based tuning is used. W1 results are in Tab. S5.

Registration. Plate-distance offsets of $\pm 0.5\,\mu\mathrm{m}$, a lateral shift by one $0.32\,\mu\mathrm{m}$ feature, and common direction-cosine offsets of 0.005 or 0.01 are applied to the optical simulation. For the updated-model distance comparison, the operator is rebuilt for the displaced plate and reconstruction parameters are reselected on validation. The learned phase itself is not retrained. Uncalibrated cases use the nominal operator and parameters. The perturbed simulations retain their nominal excitation scale. All W1 comparisons use neuron-trained PnP at $C = 8$ and are tabulated in Tab. S7.

References

[1] Thilo Balke, Fernando Davis, Cristina Garcia-Cardona, Soumendu Majee, Michael McCann, Luke Pfister, and Brendt Wohlberg. Scientific computational imaging code (SCICO). *Journal of Open Source Software*, 7(78):4722, 2022. 3

[2] Stephen Boyd, Neal Parikh, Eric Chu, Borja Peleato, and Jonathan Eckstein. Distributed optimization and statistical learning via the alternating direction method of multipliers. *Foundations and Trends in Machine Learning*, 3(1):1–122, 2011. 2

[3] Ulugbek S. Kamilov, Charles A. Bouman, Gregery T. Buzzard, and Brendt Wohlberg. Plug-and-play methods for integrating physical and learned models in computational imaging: Theory, algorithms, and applications. *IEEE Signal Processing Magazine*, 40(1):85–97, 2023. 2

[4] Soumendu Majee, Thilo Balke, Craig A. J. Kemp, Gregery T. Buzzard, and Charles A. Bouman. Multi-slice fusion for sparse-view and limited-angle 4D CT reconstruction. *IEEE Transactions on Computational Imaging*, 7:448–462, 2021. 2

[5] Gabriel Peyré and Marco Cuturi. Computational optimal transport. *Foundations and Trends in Machine Learning*, 11 (5–6):355–607, 2019. 3

[6] Leonid I. Rudin, Stanley Osher, and Emad Fatemi. Nonlinear total variation based noise removal algorithms. *Physica D: Nonlinear Phenomena*, 60(1–4):259–268, 1992. 2

[7] Yoav Shechtman, Steffen J. Sahl, Adam S. Backer, and W. E. Moerner. Optimal point spread function design for 3D imaging. *Physical Review Letters*, 113(13):133902, 2014. 10

[8] Singanallur V. Venkatakrishnan, Charles A. Bouman, and Brendt Wohlberg. Plug-and-play priors for model based reconstruction. In *IEEE Global Conference on Signal and Information Processing (GlobalSIP)*, pages 945–948, 2013. 2

[9] Kyrollos Yanny, Nick Antipa, William Liberti, Sam Dehaeck, Kristina Monakhova, Fanglin Linda Liu, Konlin Shen, Ren Ng, and Laura Waller. Miniscope3D: optimized single-shot miniature 3D fluorescence microscopy. *Light: Science & Applications*, 9:171, 2020. 10

[10] Kai Zhang, Wangmeng Zuo, Yunjin Chen, Deyu Meng, and Lei Zhang. Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising. *IEEE Transactions on Image Processing*, 26(7):3142–3155, 2017. 3